Shakti Kumar

Online Physics & Mathematics Tutor · PhD Scholar, IIT Kanpur

shaktikumarrasra@gmail.com · shaktik25@iitk.ac.in · +91 88377 08287 (WhatsApp) · linkedin.com/in/shaktikumar1310 · shaktikr.github.io/portfolio/tutoring · Kanpur, India (IST, UTC+5:30)

Profile

IIT Kanpur PhD scholar who teaches physics and mathematics through clear, step-by-step reasoning. Seven months of classroom teaching in physics, chemistry and mathematics for first-year engineering students, plus 200+ recorded worked solutions for Unacademy learners. CBSE-schooled with JEE-level problem-solving depth, adapted to the exam style of IGCSE, A-Level, IB and AP students. Teaches in English and Hindi.

Subjects Offered

Subject	Grades	Curricula and exams
Physics	9–12	CBSE, JEE Main & Advanced, IGCSE, A-Level, IB Diploma, AP Physics 1, 2 and C
Mathematics	6–12	CBSE, JEE Main & Advanced, IGCSE, A-Level, IB Diploma, AP Calculus AB/BC, SAT Math
Chemistry	9–10	CBSE, IGCSE

Teaching Experience

Graduate Teaching Assistant — IIT Kanpur · Jul 2023 – May 2024; Dec 2025 – Present

- Assists a weekly TA211 laboratory batch, keeps its attendance and marks, and supports tutorials and evaluation.

Teaching Faculty, Engineering Physics, Chemistry & Mathematics — GeeksforGurukul (ed-tech startup), PCST College, Indore · Nov 2024 – May 2025

- Taught physics, chemistry and mathematics to first-year B.Tech students through concept-based lectures and regular problem-solving sessions.

Doubt-Solving Educator (freelance, remote) — Unacademy · Aug 2021 – Jul 2022

- Recorded 200+ step-by-step video solutions to GATE Mechanical questions in engineering mathematics, thermodynamics and fluid mechanics.

Education

PhD, Materials Science & Engineering (ongoing) — IIT Kanpur · Dec 2025 – present

Coursework: Mathematics & Computational Methods; Thermodynamics of Materials; Fracture, Fatigue and Creep; AI & ML in Materials Engineering. Ongoing: Transport Phenomena; Structure & Characterization of Materials; Microscopy & Microanalysis.

M.Tech, Aerospace Engineering — IIT Kanpur · 2022 – 2024

B.Tech, Mechanical Engineering — Rajkiya Engineering College, Banda · 2018 – 2022

Class XII, CBSE — Kendriya Vidyalaya No. 1, Amritsar Cantt · 2017

Achievements

- Qualified GATE 2021 (Mechanical Engineering) in the third year of B.Tech: All India Rank 8,664 among 1,20,594 candidates (top 7.2%).
- GATE 2024 Computer Science and IT: All India Rank 22,651 among 1,23,967 candidates.
- NPTEL certifications: Speaking Effectively (IIT Kharagpur) 90%; Engineering Mechanics – Statics and Dynamics (IIT Madras) 73%; Fluid Mechanics, Elite.

Skills & Languages

- **Teaching:** concept-first lectures, step-by-step worked solutions, problem-solving sessions, recorded video explanations
- **Languages and tools:** English (professional), Hindi (native); Python and MATLAB for simulations and graphs